

Numerical integration of the Fokker–Planck equation in two dimensions

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We consider the two-dimensional Fokker–Planck equation

$$\partial_t P(x, y; t) = - \sum_{z \in \{x, y\}} \partial_z [F_z(x, y) P(x, y; t)] + \frac{D}{2} \sum_{z \in \{x, y\}} \partial_z^2 [B_z(x, y) P(x, y; t)], \quad (1)$$

where $F_z(x, y)$ and $B_z(x, y)$, with $z \in \{x, y\}$, denote the drift and diffusion functions, respectively, and the parameter $D > 0$ sets the noise intensity.

To ensure probability conservation, we rewrite Eq. (1) in conservative form,

$$\partial_t P(x, y; t) = - \sum_{z \in \{x, y\}} \partial_z J_z(x, y; t), \quad (2)$$

where the probability fluxes are defined as

$$J_z(x, y; t) = F_z(x, y) P(x, y; t) - \frac{D}{2} \partial_z [B_z(x, y) P(x, y; t)], \quad z \in \{x, y\}. \quad (3)$$

To numerically integrate Eq. (2), we employ a finite-difference discretization scheme in space combined with an alternating-direction implicit (ADI) method for time integration [1]. The spatial domain $[x_{\min}, x_{\max}] \times [y_{\min}, y_{\max}]$ is discretized using a uniform grid defined by $x_i = x_{\min} + i \Delta x$ and $y_j = y_{\min} + j \Delta y$, where $i = 0, \dots, N_x - 1$ and $j = 0, \dots, N_y - 1$, with $N_x = (x_{\max} - x_{\min})/\Delta x + 1$ and $N_y = (y_{\max} - y_{\min})/\Delta y + 1$. Time is discretized as $t_n = n \Delta t$. Introducing the notation $P_{i,j}^n = P(x_i, y_j; t_n)$, the scheme reads

$$\frac{P_{i,j}^{n+1} - P_{i,j}^n}{\Delta t} = - \frac{(\tilde{J}_x)_{i+1/2,j} - (\tilde{J}_x)_{i-1/2,j}}{\Delta x} - \frac{(\tilde{J}_y)_{i,j+1/2} - (\tilde{J}_y)_{i,j-1/2}}{\Delta y}, \quad (4)$$

where the tildes in $\tilde{J}_x$ and $\tilde{J}_y$ indicate evaluation at the appropriate temporal stage of the ADI scheme, depending on whether the corresponding term is treated implicitly or explicitly.

We discretize the numerical fluxes at the cell interfaces using centered differences as

$$(\tilde{J}_x)_{i+1/2,j} = (F_x)_{i+1/2,j} \tilde{P}_{i+1/2,j} - \frac{D}{2} \frac{(B_x)_{i+1,j} \tilde{P}_{i+1,j} - (B_x)_{i,j} \tilde{P}_{i,j}}{\Delta x}. \quad (5)$$

To enhance stability in the advective contribution, we reconstruct the interfacial probability density using a first-order upwind prescription:

$$\tilde{P}_{i+1/2,j} = [1 - (\delta_x)_{i+1/2,j}] \tilde{P}_{i+1,j} + (\delta_x)_{i+1/2,j} \tilde{P}_{i,j}, \quad (6)$$

with

$$(\delta_x)_{i+1/2,j} = \begin{cases} 1, & \text{if } (F_x)_{i+1/2,j} > 0, \\ 0, & \text{if } (F_x)_{i+1/2,j} < 0. \end{cases} \quad (7)$$

Analogous expressions to Eqs. (5) – (7) hold for the y -direction. Reflecting boundary conditions are imposed by enforcing vanishing probability fluxes at the domain boundaries, i.e., by setting $J_{-1/2,j} = J_{N_x-1/2,j} = 0 \forall j$ and $J_{i,-1/2} = J_{i,N_y-1/2} = 0 \forall i$.

For later convenience, we introduce the discrete spatial operator

$$\mathcal{L}_x[\tilde{P}_{i,j}] = -(\tilde{J}_x)_{i+1/2,j} + (\tilde{J}_x)_{i-1/2,j}, \quad (8)$$

which, upon substitution of the flux definitions, takes the tridiagonal form

$$\mathcal{L}_x[\tilde{P}_{i,j}] = (Q_x)_{i,j} \tilde{P}_{i-1,j} + (R_x)_{i,j} \tilde{P}_{i,j} + (S_x)_{i,j} \tilde{P}_{i+1,j}, \quad (9)$$

with

$$\begin{aligned} (Q_x)_{i,j} &= (F_x)_{i-1/2,j} (\delta_x)_{i-1/2,j} + \frac{D}{2} \frac{(B_x)_{i-1,j}}{\Delta x}, \\ (R_x)_{i,j} &= (F_x)_{i-1/2,j} [1 - (\delta_x)_{i-1/2,j}] - (F_x)_{i+1/2,j} (\delta_x)_{i+1/2,j} - D \frac{(B_x)_{i,j}}{\Delta x}, \\ (S_x)_{i,j} &= -(F_x)_{i+1/2,j} [1 - (\delta_x)_{i+1/2,j}] + \frac{D}{2} \frac{(B_x)_{i+1,j}}{\Delta x}. \end{aligned} \quad (10)$$

Analogous definitions apply both to the operator $\mathcal{L}_y$ and to the coefficients Q_y , R_y and S_y .

In terms of the operators $\mathcal{L}_x$ and $\mathcal{L}_y$, Eq. (4) becomes

$$P_{i,j}^{n+1} - P_{i,j}^n = \nu_x \mathcal{L}_x[\tilde{P}_{i,j}] + \nu_y \mathcal{L}_y[\tilde{P}_{i,j}], \quad (11)$$

where we have introduced the parameters $\nu_x = \Delta t / \Delta x$ and $\nu_y = \Delta t / \Delta y$.

Time integration of Eq. (11) is performed using an ADI scheme consisting of two successive half-steps of duration $\Delta t/2$:

- In the first half-step, the x -direction is treated implicitly while the y -direction is treated explicitly:

$$P_{i,j}^{n+1/2} - P_{i,j}^n = \frac{\nu_x}{2} \mathcal{L}_x[P_{i,j}^{n+1/2}] + \frac{\nu_y}{2} \mathcal{L}_y[P_{i,j}^n], \quad (12)$$

which can be rearranged as

$$\left(1 - \frac{\nu_x}{2} \mathcal{L}_x\right) P_{i,j}^{n+1/2} = \left(1 + \frac{\nu_y}{2} \mathcal{L}_y\right) P_{i,j}^n. \quad (13)$$

For fixed j , this equation defines a tridiagonal linear system for the unknowns $P_{i,j}^{n+1/2}$, which can be solved using standard methods such as the Thomas algorithm [1].

- In the second half-step, the implicit and explicit roles of the spatial directions are exchanged:

$$P_{i,j}^{n+1} - P_{i,j}^{n+1/2} = \frac{\nu_x}{2} \mathcal{L}_x[P_{i,j}^{n+1/2}] + \frac{\nu_y}{2} \mathcal{L}_y[P_{i,j}^{n+1}], \quad (14)$$

leading to

$$\left(1 - \frac{\nu_y}{2} \mathcal{L}_y\right) P_{i,j}^{n+1} = \left(1 + \frac{\nu_x}{2} \mathcal{L}_x\right) P_{i,j}^{n+1/2}. \quad (15)$$

For fixed i , this equation yields a tridiagonal linear system of equations, now for the unknowns $P_{i,j}^{n+1}$.

The decomposition of time integration in alternating-implicit direction substeps substantially reduces the computational cost while preserving the stability advantages of an implicit integration scheme.

References

- [1] Keith W Morton and David Francis Mayers. *Numerical Solution of Partial Differential Equations: An Introduction*. Cambridge University Press, 2nd edition, 2005.